

6 $p^3 \times p^x = p^9$

(a) Find the value of x .

$$x =$$

(1)

$$(7^2)^y = 7^{10}$$

(b) Find the value of y .

$$y =$$

(1)

$100^a \times 1000^b$ can be written in the form 10^w

(c) Show that $w = 2a + 3b$

(2)

(Total for Question 6 is 4 marks)